

THE FIVE-BIT THRESHOLD

Theorem: Under natural encodings, $K(C) \geq 5$ bits for universal C

$$\text{Logic}(2) + \text{Memory}(1) + \text{Control}(1) + \text{State}(1) = 5 \text{ bits}$$

Tight: Tag(2,2), SK calculus, Rule 110 achieve exactly 5 bits

Logic ≥ 2

Post 1941

Memory ≥ 1

Turing 1936

Control ≥ 1

Böhm-Jacopini

State ≥ 1

Minsky 1967

Conjecture: The threshold reflects the cost of encoding CONTROL

The capacity for conditional, context-dependent state transitions